
EMBEDDED HACKING

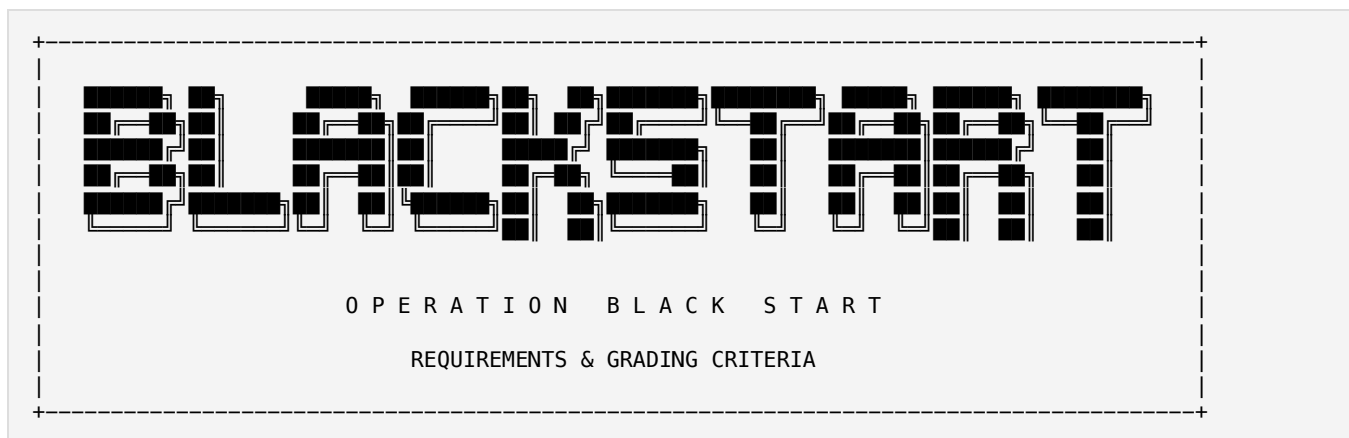
Reverse Engineering the Raspberry Pi Pico 2

Operation Black Start - Requirements & Grading Criteria

ARM Cortex-M33 • GDB • Ghidra

September 25, 2026

For educational and defensive security research only



Project Overview

Students are the reverse-engineering reserve team called in after WorldGrid Compact's emergency firmware build shipped a miscompiled safety threshold and a false status string to its GRID-7 relay fleet. Students reverse engineer `CTF-01.bin` with Ghidra, locate two real defects, patch them in the binary, export a corrected image, flash it to real hardware, and prove the corrected behavior with the debugger and the console.

The challenge is separate from all FINAL projects and contains no FINAL-project answer, constant, address, bug, or patch.

Learning Objectives

- Decode an ARM Cortex-M33 vector table and identify the reset handler and initial stack pointer.
- Translate Thumb reset-vector addresses into real function entry points.
- Locate a miscompiled boundary comparison and reason about its immediate value.
- Patch compare instructions and a status string in a raw binary with Ghidra.
- Export and UF2-convert a corrected image, then verify it on real hardware.
- Capture derived state with GDB and read UART console output.

Students must use only Weeks 1-3 concepts: ARM registers and stack behavior, UART output, GDB, Ghidra static analysis and binary patching, vector tables, reset startup, XIP, and Thumb addressing.

Deliverables Checklist

#	Deliverable	Format	Criterion
1	Ghidra project screenshot (project name, processor, base address)	PNG/JPG	1.1
2	<code>main()</code> and status-loop addresses	Inside <code>CTF-01-Answers.md</code>	1.2
3	Vector table base, initial SP, reset pointer	Inside <code>CTF-01-Answers.md</code>	1.3
4	Thumb bit explanation	Inside <code>CTF-01-Answers.md</code>	1.4
5	Bug #1 evidence and patches (both compare sites)	Inside <code>CTF-01-Answers.md</code>	2.1-2.6

#	Deliverable	Format	Criterion
6	Bug #2 evidence and patch (six characters)	Inside CTF-01-Answers.md	3.1-3.4
7	Recovered dispatch frame and address	Inside CTF-01-Answers.md	4.1-4.2
8	CTF-01_fixed.bin	BIN file	5.1
9	CTF-01_fixed.uf2	UF2 file	5.2
10	Corrected console transcript	Inside CTF-01-Answers.md	5.3
11	Summary table of all patches	Inside CTF-01-Answers.md	5.4
12	Written reflection	Inside CTF-01-Answers.md	6.1-6.2

Required Tools and Equipment

Tool	Purpose
Raspberry Pi Pico 2	Isolated target
3.3 V USB-UART adapter	UART capture on GPIO 0 (TX) / GPIO 1 (RX)
Serial monitor	Observe output
Ghidra	Static analysis and binary patching
Python (uf2conv.py)	UF2 conversion
CTF-01.bin and CTF-01.uf2	Supplied artifacts

UART settings: **115200 baud, 8 data bits, no parity, 1 stop bit.**

Artifact Identity

The instructor-issued artifact hashes are:

```
CTF-01.bin  6FD296F7A85F243FB26BF6BFFCBEAB26815FD8915101A81F72069063A5635E5A
CTF-01.uf2  980F04369C23AD32A063DFE18DE5AF08DF3830138FC7E7898B1F011B4F5E1D9D
```

Grading Rubric - Detailed Breakdown

Task 1: Setup and Initial Analysis (15 points)

Criterion	Points	Full credit	Partial credit	No credit
Criterion 1.1: Ghidra Project Setup	3	Correct project name, ARM Cortex 32-bit little endian, base 0x10000000	One item off	Not set up

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Criterion	Points	Full credit	Partial credit	No credit
Criterion 1.2: main() and Status-Loop Addresses	4	Both addresses correct	One correct	Neither found
Criterion 1.3: Vector Table Decoding	4	Correct base, initial SP, reset pointer	One missing	Not found
Criterion 1.4: Thumb Addressing	4	Correctly clears bit 0 and identifies <code>main()</code>	General explanation	Incorrect

Task 2: Find and Patch Bug #1: The Miscalibrated Safety Threshold (30 points)

Criterion	Points	Full credit	Partial credit	No credit
Criterion 2.1: Locate Compare Site A	5	Correct address and original bytes	Address off	Not found
Criterion 2.2: Locate Compare Site B	5	Correct address and original bytes	Address off	Not found
Criterion 2.3: Correct Immediate-Value Reasoning	8	Explains the <code><</code> to <code><=</code> transform and gives <code>0x3B</code>	Correct value, no reasoning	Wrong value
Criterion 2.4: Patch Compare Site A	4	Byte change verified	Wrong byte	Not patched
Criterion 2.5: Patch Compare Site B	4	Byte change verified	Wrong byte	Not patched
Criterion 2.6: Explain Why Both Sites Must Be Patched	4	Clear explanation of the two independent comparisons	Vague	Missing

Task 3: Find and Patch Bug #2: The False Signal Banner (20 points)

Criterion	Points	Full credit	Partial credit	No credit
Criterion 3.1: Locate the Banner String	5	Correct address	Approximate	Not found
Criterion 3.2: Patch Six Characters	8	All six bytes changed, length preserved	Correct text, wrong bytes documented	Wrong length
Criterion 3.3: Character-by-Character Documentation	4	Original vs patched byte for all six characters	Partial	Missing
Criterion 3.4: Explain the Danger of a Hardcoded Status Word	3	Clear, specific reasoning	Generic	Missing

Task 4: Recover the Quarantined Dispatch Frame (10 points)

Criterion	Points	Full credit	Partial credit	No credit
Criterion 4.1: Recover the Dispatch Frame	6	Correct address and full text	Partial text	Not found
Criterion 4.2: Explain Why It Is Never Transmitted	4	Clear static-analysis explanation	Vague	Missing

Task 5: Export and Verify (20 points)

Criterion	Points	Full credit	Partial credit	No credit
Criterion 5.1: Export CTF-01_fixed.bin	4	Valid patched binary	Corrupted	Not submitted
Criterion 5.2: Convert to CTF-01_fixed.uf2	4	Correct base and family flags	Wrong flags	Not submitted
Criterion 5.3: Hardware Verification	8	Corrected console output confirmed (CRITICAL/HELD in 1s stream; DANGER at boot/Ghidra)	Some lines corrected	No verification
Criterion 5.4: Summary Table of All Patches	4	Complete address and before/after table	Missing entries	No table

Task 6: Written Reflection (5 points)

Criterion	Points	Full credit	Partial credit	No credit
Criterion 6.1: "Rushed Build" Is Not an Excuse	2	Specific, grounded reasoning	Generic	Missing
Criterion 6.2: One Engineering Practice per Bug	3	Concrete practice for each bug	One bug only	Missing

Common Pitfalls

Pitfall	Consequence	Avoidance
Patching only one threshold site	One status line still lies	Patch both <code>0x100001FC</code> and <code>0x1000020A</code>
Assuming the immediate equals the limit	Off-by-one, wrong boundary	Use <code>0x3B</code> (59), not <code>0x3C</code> (60)
Using Patch Instruction before IT block	Re-disassembler context conflict swallows Site B	In Listing press <code>C</code> -> edit byte in Bytes window (pencil) -> press <code>D</code>
Missing boot banner in serial terminal	PuTTY misses one-time 5ms boot banner	Pulse RUN to GND while connected to capture
Replacing a string with a different length	Corrupts adjacent flash	<code>NORMAL</code> and <code>DANGER</code> are both 6 bytes
Treating an odd vector address as invalid	Thumb analysis fails	Clear bit 0
Modifying the quarantined dispatch frame	Destroys evidence	Recover it, do not patch it

How To Breadboard

- **Raspberry Pi Pico 2** powered over USB.

- **3.3 V USB-UART adapter:**
 - Adapter RX to Pico GP0 (UARTo TX)
 - Adapter TX to Pico GP1 (UARTo RX)
 - Adapter GND to Pico GND
 - Do not connect the adapter VCC while the Pico is USB powered.
 - **Serial monitor:** 115200 baud, 8 data bits, no parity, 1 stop bit.
 - No other peripherals are required; all evidence is obtained from the console.
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Memory Map Reference

Region	Address	Purpose
Bootrom	0x00000000	Immutable boot code
Flash/XIP	0x10000000	Vector table, code, constants, strings
SRAM	0x20000000	Stack and writable state

Deadline & Submission

- Create a folder containing the Ghidra screenshot, `CTF-01_fixed.bin`, and `CTF-01_fixed.uf2`.
 - Write all written answers in a single file named `CTF-01-Answers.md` inside that folder.
 - ZIP the folder as `lastname-firstname-CTF-01.zip`.
 - Submit the ZIP before the posted deadline; late submissions lose 10 percent per day.
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Grade Scale

Grade	Percentage	Points
A+	97-100%	97-100
A	93-96%	93-96
A-	90-92%	90-92
B+	87-89%	87-89
B	84-86%	84-86
B-	80-83%	80-83
C	70-79%	70-79
F	0-69%	0-69

Academic Integrity

Use only the supplied Pico 2 and firmware. Do not connect the exercise to an operational grid, water plant, public network, military system, or third-party device. This is a controlled, isolated educational exercise. All analysis and patches must be your own work; sharing binaries, addresses, or answers is a violation of the academic integrity policy.

Reference Material

Topic	Reference
ARM Cortex-M33 registers and stack	Week 1
UART output and console capture	Week 2
Vector tables, reset startup, and XIP	Week 2
Ghidra static analysis and binary patching	Week 3
Thumb addressing	Week 3